

PRIME VALUES OF DIGITAL FUNCTIONS ALONG THE PRIMES

ABSTRACT. Let g be an integer valued strongly b -additive function with $\gcd(g(1), \dots, g(b-1)) = 1$ and with nonnegative mean μ_g over the digits. We show that $g(p)$ is prime for infinitely many primes p . When $\mu_g > 0$ we prove a Mertens type formula: the sum of $1/p$ over the primes $p < X$ with $g(p)$ prime is $\frac{d_g}{\varphi(d_g)} \log_3 X + C_{g,1} + O(1/\log \log X)$, where $d_g \mid b-1$ is a modulus attached to g and $C_{g,1}$ is a constant. The analogous formula holds for the primes whose first j iterates under g are prime, and for the base b digit sum it sharpens theorems of Harman. We show further that the number of $p \leq x$ with $g(p)$ prime is $\ll \pi(x)/\log \log x$, and that it is of that exact order on a large set of x ; under a hypothesis on primes in intervals of length $\sqrt{y}$, it is of that order at every x . The normal order of $\omega(g(p))$ is $\log_3 p$. The proofs rest on a local limit theorem of Martin, Mauduit and Rivat. Our running example is the sum S of the squares of the decimal digits: the primes p with $S(p)$ prime, sequence A052034 of the OEIS, are infinite in number.

1. INTRODUCTION

Fix an integer $b \geq 2$. A function $g: \mathbb{N} \rightarrow \mathbb{Z}$ is *strongly b -additive* if

$$g\left(\sum_{j \geq 0} \varepsilon_j b^j\right) = \sum_{j \geq 0} g(\varepsilon_j) \quad (0 \leq \varepsilon_j \leq b-1),$$

so that g is determined by the values $g(0) = 0, g(1), \dots, g(b-1)$, whose mean and variance we denote by

$$\mu_g = \frac{1}{b} \sum_{a=0}^{b-1} g(a), \quad \sigma_g^2 = \frac{1}{b} \sum_{a=0}^{b-1} (g(a) - \mu_g)^2.$$

Write $\log_1 = \log$ and $\log_i = \log \circ \log_{i-1}$. We prove that if $\gcd(g(1), \dots, g(b-1)) = 1$ and $\mu_g \geq 0$, then $g(p)$ is prime for infinitely many primes p . If in addition $\mu_g > 0$, then the sum of $1/p$ over the primes $p < X$ with $g(p)$ prime equals $\frac{d_g}{\varphi(d_g)} \log_3 X + C_{g,1} + O(1/\log_2 X)$, for a constant $C_{g,1}$ and a divisor d_g of $b-1$ attached to g in (4). We state these as Theorems 1.1 and 1.3, the latter in an iterated form.

Our running example is $b = 10$ and

$$S(n) = \sum_{j \geq 0} \varepsilon_j(n)^2, \quad n = \sum_{j \geq 0} \varepsilon_j(n) 10^j, \quad 0 \leq \varepsilon_j(n) \leq 9,$$

the sum of the squares of the decimal digits, for which $d_S = 1$. The primes p for which $S(p)$ is again prime begin

$$11, 23, 41, 61, 83, 101, 113, 131, 137, 173, 179, 191, \dots$$

and were recorded by De Geest in 1999 as sequence A052034 of the OEIS [13], where the citations are to the recreational literature. We have not located anywhere a proof, or an assertion, that the sequence is infinite. It is.

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In a paper of 1968, Gel'fond [5] raised several questions about the interaction between the digits of an integer and its multiplicative structure, among them the distribution in arithmetic progressions of the decimal digit sum $s(p)$ as p runs over the primes. The question stood for four decades. It was answered by Mauduit and Rivat [12], who proved that $s(p)$ is equidistributed modulo m for every m coprime to 9, with a saving of a power of x in the error term. Shortly before, Drmota, Mauduit and Rivat [3] had obtained a local version for the base q digit sum s_q . Writing $\mu_q = (q - 1)/2$, $\sigma_q^2 = (q^2 - 1)/12$ and

$$\delta_q(k, x) = \frac{k - \mu_q \log_q x}{\sigma_q \sqrt{\log_q x}},$$

their result reads

$$\#\{p \leq x : s_q(p) = k\} = \frac{q-1}{\varphi(q-1)} \cdot \frac{\pi(x)}{\sqrt{2\pi\sigma_q^2 \log_q x}} \left(e^{-\delta_q(k,x)^2/2} + O((\log x)^{-\frac{1}{2}+\varepsilon}) \right), \quad (1)$$

uniformly in integers $k \geq 0$ with $\gcd(k, q-1) = 1$; the coprimality condition cannot be dropped, since $s_q(n) \equiv n \pmod{q-1}$. Harman [6] observed that (1) settles a question much easier to ask than to answer: every sufficiently large k with $\gcd(k, q-1) = 1$ is the digit sum of some prime, and prime values of k are available in abundance, so there are infinitely many primes whose digit sum is again prime. He proved a good deal more: a Mertens type formula for the reciprocals of those primes, and an iterated version for primes whose digit sum is prime, whose digit sum is prime, and so on. On a hypothesis about primes in short intervals he obtained an asymptotic formula for their number.

The role of (1) is played here by a local limit theorem of Martin, Mauduit and Rivat [11], quoted as Theorem 2.1 in Section 2; everything below is deduced from it. Implied constants may always depend on the base b and on the weight g ; further dependence will be noted explicitly.

Theorem 1.1. *Let $b \geq 2$ and let g be strongly b -additive with integer values, $\gcd(g(1), \dots, g(b-1)) = 1$ and $\mu_g \geq 0$. Then $g(p)$ is prime for infinitely many primes p . More precisely, (i) if $\mu_g > 0$, then for every sufficiently large prime k there are at least $x_k(\log x_k)^{-2}$ primes $p \leq x_k$ with $g(p) = k$, where $x_k = b^{k/\mu_g}$; and (ii) if $\mu_g = 0$, then for every integer k coprime to $b-1$, and in fact to the divisor d_g of $b-1$ defined in Section 2, there are $\gg \pi(x)(\log x)^{-1/2}$ primes $p \leq x$ with $g(p) = k$ once x is large in terms of k .*

Theorem 1.2. *Assume the hypotheses of Theorem 1.1. If $\mu_g > 0$ there is a constant $C = C(b, g)$ with*

$$\#\{p \leq x : g(p) \text{ prime}\} \geq x \exp(-C(\log x)^{0.525})$$

for all large x , while if $\mu_g = 0$ the count is $\gg \pi(x)(\log x)^{-1/2}$, a single prime target sufficing.

Theorem 1.2 is not of the right order, the truth being of order $\pi(x)/\log \log x$ by Theorem 1.8 below, but it has the merit of holding at every large x , which Theorem 1.8 does not. The reciprocal sum, on the other hand, can be evaluated exactly, and in an iterated form. Let $\mathcal{P}^{(0)}$ be the set of all primes, and put

$$\mathcal{P}^{(j)} = \{p \text{ prime} : g(p) \in \mathcal{P}^{(j-1)}\} \quad (j \geq 1),$$

so that $\mathcal{P}^{(1)}$ consists of the primes at which g takes a prime value, $\mathcal{P}^{(2)}$ of those for which $g(p)$ is in turn such a prime, and so on.

Theorem 1.3. *Let b and g be as in Theorem 1.1 with $\mu_g > 0$, and let $j \geq 1$ be fixed. There is a constant $C_{g,j}$ such that*

$$\sum_{\substack{p < X \\ p \in \mathcal{P}^{(j)}}} \frac{1}{p} = \left(\frac{d_g}{\varphi(d_g)} \right)^j \log_{j+2} X + C_{g,j} + O\left(\frac{1}{\log_{j+1} X} \right),$$

where $d_g \mid b-1$ is defined in (4), and $C_{g,j}$ and the implied constant depend also on j .

The remainder is dictated by the shape of the proof, which passes through (8) evaluated at $Y \asymp \log X$, so that each iteration costs a logarithm. For $j = 1$ the argument gives more, namely

$$\sum_{\substack{p < X \\ g(p) \text{ prime}}} \frac{1}{p} = \frac{d_g}{\varphi(d_g)} \log(\log_2 X + \kappa_g) + C_{g,1} + O(e^{-c\sqrt{\log_2 X}}), \quad \kappa_g = \log \frac{\mu_g}{\log b}, \quad (2)$$

of which the displayed formula is the crude form, since $\log(\log_2 X + \kappa_g) = \log_3 X + O(1/\log_2 X)$.

For $g = s_b$ one has $d_g = b-1$, and the leading term is due to Harman [6, Theorems 2 and 3], who states the remainder as $O(1)$; that the difference between the sum and its main term tends to a limit is the new information. For $b = 10$ and $g = S$ the leading coefficient is 1.

It is natural to ask why the reciprocal sum is accessible when the asymptotic count is not. For a single x one would have to count the primes k in a window of length $\asymp \sqrt{y}$ about $y = \mu_g \log_b x$; this is the obstruction described after Conjecture 1.7. Partial summation replaces that count by an integral over $t \leq X$, during which the centre $\mu_g \log_b t$ sweeps a long range. Reversing the order of summation then asks only for the primes k up to $\mu_g \log_b X$, which is Mertens' theorem. The averaging in t does the work that a short interval hypothesis would otherwise have to do. Theorem 1.3 may accordingly be read as saying that Conjecture 1.7 holds on average.

Lagrange interpolation attaches to g a unique $P \in \mathbb{Q}[X]$ with $\deg P \leq b-1$, $P(0) = 0$ and $P(a) = g(a)$ for $0 \leq a < b$, and conversely every such P that is integer valued on the digits arises in this way. Theorem 1.1 is therefore a statement about digit weights defined by polynomials, and about all of them, its hypotheses reading

$$\gcd(P(1), \dots, P(b-1)) = 1, \quad \frac{1}{b} \sum_{a=0}^{b-1} P(a) \geq 0.$$

For a fixed P and large b , Faulhaber's formula settles the second condition by the sign of the leading coefficient of P . Integrality of the coefficients is not needed, only of the values. For $P(X) = \binom{X}{2}$ in base ten one has $\sum_j \binom{\varepsilon_j(n)}{2} = \frac{1}{2}(S(n) - s(n))$, whose digit values 0, 0, 1, 3, 6, 10, 15, 21, 28, 36 have greatest common divisor 1 and mean 12, so $\frac{1}{2}(S(p) - s(p))$ is prime for infinitely many primes p . We record three consequences.

Corollary 1.4. *Let $b \geq 2$ and $r \geq 1$. There are infinitely many primes p such that the sum of the r th powers of the base b digits of p is prime.*

Corollary 1.5. *Let $b \geq 2$ and let $c \in \{1, \dots, b-1\}$. There are infinitely many primes in whose base b expansion the digit c occurs a prime number of times.*

Corollary 1.6. *Let b and g be as in Theorem 1.1 with $\mu_g > 0$, and let $t \geq 1$. Write $g^{(i)}$ for the i th iterate of g . There are infinitely many primes p such that $g(p), g^{(2)}(p), \dots, g^{(t)}(p)$ are all prime.*

Corollary 1.4 with $r = 2$ and $b = 10$ is the assertion about A052034 made above; Corollary 1.6 follows on applying clause (i) of Theorem 1.1 t times in succession. For $g = s$ the last is

contained, in a stronger and quantitative form, in Harman [6]; Lehmann [8] has since made the underlying surjectivity explicit and deduced the existence of infinitely many primes whose whole chain of iterated digit sums consists of primes.

The exponent 0.525 in Theorem 1.2 comes from the theorem of Baker, Harman and Pintz [1] on primes in short intervals, and any improvement there improves it. The truth should be much larger. Heuristically the values $g(p)$ for $p \leq x$ cluster around $\mu_g \log_b x$ in a range of length of order $\sqrt{\log x}$, and a prime should be about as likely to occur there as at a random integer of that size; this suggests the following.

Conjecture 1.7. *Let b and g be as in Theorem 1.1 with $\mu_g > 0$. Then*

$$\#\{p \leq x : g(p) \text{ prime}\} \sim \frac{d_g}{\varphi(d_g)} \cdot \frac{\pi(x)}{\log(\mu_g \log_b x)} \quad (x \rightarrow \infty);$$

for $b = 10$ and $g = S$ the right side is $\pi(x)/\log(\frac{57}{2} \log_{10} x)$.

Conjecture 1.7 seems out of reach at present, for a reason that has nothing to do with digits. Even granting the full strength of Theorem 2.1, an asymptotic count requires the primes to be distributed as expected in intervals of length of order $\sqrt{y}$ about $y = \mu_g \log_b x$, and no such statement is known; as Harman notes for s , this is not obtainable even on the Riemann Hypothesis together with the pair correlation conjecture. Nor is the length $\sqrt{y}$ an artifact: it is the length of the range the values $g(p)$ occupy, and Theorem 2.1 says nothing about which integers there are prime. The conjecture can nevertheless be bracketed unconditionally: from above at every x , and from below on a large set of x .

Theorem 1.8. *Let b and g be as in Theorem 1.1 with $\mu_g > 0$, and write $M(x) = \#\{p \leq x : g(p) \text{ prime}\}$. Then*

$$M(x) \ll \frac{\pi(x)}{\log \log x} \quad (x \geq 16),$$

and there are $\delta > 0$ and $t_0 \geq 16$ such that, for all large X , the set of $t \in [t_0, X]$ with

$$M(t) \geq \delta \frac{\pi(t)}{\log \log t}$$

has measure at least $\delta \log_3 X$ with respect to $\nu = dt/(t \log t \log_2 t)$, which assigns mass $\log_3 X + O(1)$ to $[t_0, X]$.

So the order of magnitude in Conjecture 1.7 is correct, and is attained on a set of positive density for ν , that is, on the $\log_3$ scale. What is missing is that it is attained at *every* large x , and that is where a hypothesis on primes in short intervals enters.

The same window controls the factorisation of $g(p)$ as well as its primality; ω and Ω denote the number of prime divisors of an integer, without and with multiplicity.

Theorem 1.9. *Let b and g be as in Theorem 1.1 with $\mu_g > 0$. For all but $o(\pi(x))$ of the primes $p \leq x$,*

$$\omega(g(p)) = (1 + o(1)) \log_3 p, \quad \Omega(g(p)) = (1 + o(1)) \log_3 p.$$

The normal order of $\omega(g(p))$ is thus the same as that of $\omega(n)$ for n of size $\log p$; Section 6.4 records what can instead be arranged for the smallest prime factor of $g(p)$, a statement about rare p rather than about the typical one. Carrying the same moment computation to the r th order, with the cut-off $z = y^{1/(2r)}$ in place of $y^{1/4}$, gives the Erdős–Kac law for $\omega(g(p))$.

The hypothesis that would settle Conjecture 1.7 can be named precisely.

Hypothesis 1.10 (SI). *For every fixed $c > 0$, $\pi(y + c\sqrt{y}) - \pi(y) \sim c\sqrt{y}/\log y$ as $y \rightarrow \infty$.*

Theorem 1.11. *Let b and g be as in Theorem 1.1 with $\mu_g > 0$. On Hypothesis 1.10,*

$$\#\{p \leq x : g(p) \text{ prime}\} \sim \frac{d_g}{\varphi(d_g)} \cdot \frac{\pi(x)}{\log(\mu_g \log_b x)} \quad (x \rightarrow \infty).$$

In particular Conjecture 1.7 holds on Hypothesis 1.10.

Harman [6, Theorem 6] proves the corresponding statement for s on a hypothesis of this kind, in his case about intervals shorter than $\sqrt{y}$, which implies Hypothesis 1.10.

2. A LOCAL LIMIT THEOREM FOR DIGITAL FUNCTIONS

Let $\mathcal{F}_+$ denote the set of strongly b -additive functions $g: \mathbb{N} \rightarrow \mathbb{Z}$ satisfying

$$\gcd(g(1), \dots, g(b-1)) = 1, \quad (3)$$

and for $g \in \mathcal{F}_+$ set

$$d_g = \gcd(g(2) - 2g(1), g(3) - 3g(1), \dots, g(b-1) - (b-1)g(1), b-1). \quad (4)$$

The integer d_g divides $b-1$ and records the congruence rigidity of g . Indeed $b \equiv 1 \pmod{d_g}$ and $g(a) \equiv ag(1) \pmod{d_g}$ for each digit a , so that

$$g(n) \equiv g(1) \sum_{j \geq 0} \varepsilon_j(n) \equiv g(1) n \pmod{d_g} \quad (n \geq 1);$$

the value $g(p)$ is thus confined modulo d_g to a class determined by p .

Theorem 2.1 (Martin, Mauduit, Rivat [11, Théorème 1]). *Let $b \geq 2$ and $g \in \mathcal{F}_+$, and fix $\varepsilon \in (0, \frac{1}{2})$. Uniformly for $x > 2$ and $k \in \mathbb{Z}$,*

$$\#\{p \leq x : g(p) = k\} = \frac{d_g \pi_k(x)}{\sqrt{2\pi\sigma_g^2 \log_b x}} \exp\left(-\frac{(k - \mu_g \log_b x)^2}{2\sigma_g^2 \log_b x}\right) + O\left(\frac{\pi(x)}{(\log x)^{1-\varepsilon}}\right), \quad (5)$$

where $\pi_k(x) = \#\{p \leq x : g(1)p \equiv k \pmod{d_g}\}$ and the implied constant depends also on ε .

Martin, Mauduit and Rivat prove the more general statement in which the left side of (5) carries a factor $e(\beta p)$, uniformly in real β ; the case $\beta = 0$ is what we need. The only consequence they draw is that (βp) is equidistributed modulo 1, for irrational β , as p runs over the primes with $g(p) = \lfloor \mu_g \log_b p \rfloor$; prime values of g are not discussed. Their earlier papers [9, 10] supply the underlying exponential sum estimates; Bassily and Kátai [2] had established the central limit theorem, without the local refinement, for strongly q -additive functions along polynomial sequences.

Theorem 2.1 contains (1): for $g = s_b$ one has $g(a) = a$, so every difference $g(a) - ag(1)$ vanishes, $d_g = b-1$, and $\pi_k(x) = \pi(x; k, b-1) \sim \pi(x)/\varphi(b-1)$ when $\gcd(k, b-1) = 1$. Substituting into (5) returns (1) with $q = b$, coprimality condition and constant $(b-1)/\varphi(b-1)$ included.

We record two elementary consequences of (3).

Lemma 2.2. *Let $g \in \mathcal{F}_+$ with $b \geq 2$. Then $\sigma_g > 0$ and $\gcd(g(1), d_g) = 1$.*

Proof. If $\sigma_g = 0$ then g is constant on $\{0, 1, \dots, b-1\}$, and since $g(0) = 0$ that constant is 0, contradicting (3). For the second assertion, suppose a prime ℓ divides both $g(1)$ and d_g . By (4), ℓ divides $g(a) - ag(1)$ for $2 \leq a \leq b-1$, whence $\ell \mid g(a)$ for all a , again contradicting (3). $\square$

3. DIGITAL FUNCTIONS TAKE PRIME VALUES: PROOFS OF THEOREMS 1.1 AND 1.2

Proof of Theorem 1.1. The proof is a direct application of Theorem 2.1. Fix $\varepsilon = \frac{1}{4}$ there, and let k be an integer with $\gcd(k, d_g) = 1$, which holds in particular when $\gcd(k, b-1) = 1$. By Lemma 2.2 the residue $g(1)$ is invertible modulo d_g ; so the congruence $g(1)p \equiv k \pmod{d_g}$ confines p to the single class $c \equiv g(1)^{-1}k$, which is coprime to d_g . The prime number theorem for arithmetic progressions therefore gives

$$\pi_k(x) = \frac{\pi(x)}{\varphi(d_g)} (1 + o(1)) \quad (x \rightarrow \infty), \quad (6)$$

the modulus d_g being fixed.

Suppose first that $\mu_g > 0$, and let k be a prime exceeding $b-1$. Choose the range of summation to suit the target:

$$x_k = b^{k/\mu_g}, \quad \text{so that} \quad \mu_g \log_b x_k = k$$

and the exponential in (5) equals 1. With this choice, (5) and (6) give

$$\#\{p \leq x_k : g(p) = k\} = \frac{d_g}{\varphi(d_g)} \cdot \frac{\pi(x_k)}{\sqrt{2\pi\sigma_g^2 \log_b x_k}} (1 + o(1)) + O\left(\frac{\pi(x_k)}{(\log x_k)^{3/4}}\right).$$

The main term has order $\pi(x_k)(\log x_k)^{-1/2}$ and so dominates; consequently

$$\#\{p \leq x_k : g(p) = k\} \gg \frac{x_k}{(\log x_k)^{3/2}} \quad (7)$$

once k is large; in particular the right side exceeds $x_k(\log x_k)^{-2}$, which is the form stated in Theorem 1.1. The count being positive, some prime p has $g(p) = k$. As k runs through the primes, $x_k \rightarrow \infty$, and the primes so produced cannot repeat: distinct k give distinct values $g(p)$.

Suppose now that $\mu_g = 0$ and that k is fixed with $\gcd(k, b-1) = 1$. The exponential in (5) is $\exp(-k^2/(2\sigma_g^2 \log_b x)) = 1 + o(1)$, so (5) and (6) give

$$\#\{p \leq x : g(p) = k\} = \frac{d_g}{\varphi(d_g)} \cdot \frac{\pi(x)}{\sqrt{2\pi\sigma_g^2 \log_b x}} (1 + o(1)) + O\left(\frac{\pi(x)}{(\log x)^{3/4}}\right) \gg \frac{\pi(x)}{\sqrt{\log x}}.$$

Taking for k a prime coprime to $b-1$, of which there are infinitely many, finishes the proof. $\square$

Note the direction of the argument: rather than seek a prime inside a prescribed short interval, which would be hopeless, one selects the prime target first and then slides the range x until the distribution of $g(p)$ is centred on it.

Proof of Theorem 1.2. This time the range is prescribed and the target must be fitted to it. Let x be large, put $y = \mu_g \log_b x$, and let k be the largest prime with $k \leq y$. By the theorem of Baker, Harman and Pintz [1], the interval $[y - y^{0.525}, y]$ contains a prime once y is large, so $y - k \ll y^{0.525} \ll (\log x)^{0.525}$. Hence

$$x_k = b^{k/\mu_g} = x \cdot b^{-(y-k)/\mu_g} \geq x \exp(-C_1(\log x)^{0.525})$$

for a suitable $C_1 = C_1(b, g)$, and in particular $x_k \leq x$. Every prime counted by (7) satisfies $p \leq x_k \leq x$ and $g(p) = k$ prime, so

$$\#\{p \leq x : g(p) \text{ prime}\} \geq \#\{p \leq x_k : g(p) = k\} \gg \frac{x_k}{(\log x_k)^{3/2}} \geq x \exp(-C(\log x)^{0.525})$$

with $C = C_1 + 1$, say, for all large x . The final assertion is clause (ii) of Theorem 1.1 applied to a single prime k coprime to $b - 1$. $\square$

Proofs of Corollaries 1.4 and 1.5. In both cases the function g is strongly b -additive by construction. For Corollary 1.4, $g(a) = a^r$ has $g(1) = 1$, so (3) holds, and $\mu_g = \frac{1}{b} \sum_{a < b} a^r > 0$. For Corollary 1.5, $g(a) = 1$ if $a = c$ and $g(a) = 0$ otherwise, so (3) holds and $\mu_g = 1/b > 0$. Theorem 1.1 applies in each case. $\square$

Proof of Corollary 1.6. The set in question is $\mathcal{P}^{(t)}$, whose reciprocal sum diverges by Theorem 1.3. Alternatively, and without Theorem 1.3: let k_t be a prime large enough for clause (i) of Theorem 1.1 and construct $k_{t-1}, \dots, k_0$ downwards, clause (i) supplying at each step a prime k_i with $g(k_i) = k_{i+1}$. Since $g(n) \ll \log n$, the relation $k_{i+1} = g(k_i)$ forces $k_i \gg \exp(ck_{i+1})$ for some $c = c(b, g) > 0$, so each k_i in turn is large enough for the next application, and $p = k_0$ works. $\square$

4. A MERTENS FORMULA FOR THESE PRIMES: PROOF OF THEOREM 1.3

We first record the large deviation estimate that lets us discard the primes at which g is far from its mean. Let $R = \max_{a < b} g(a) - \min_{a < b} g(a)$, which is positive by Lemma 2.2.

Lemma 4.1. *Let g be strongly b -additive with $\mu_g \geq 0$. For $t \geq b$ and $u > 0$,*

$$\#\{n \leq t : |g(n) - \mu_g \log_b t| \geq u + \mu_g\} \leq 2bt \exp\left(-\frac{u^2}{2R^2(\log_b t + 1)}\right).$$

Proof. The lemma is an instance of Hoeffding's inequality. Put $m = \lfloor \log_b t \rfloor + 1$, so that $b^m > t$ and every $n \leq t$ lies in $[0, b^m)$. Under the uniform measure on that range the digits $\varepsilon_0, \dots, \varepsilon_{m-1}$ are independent and uniform on $\{0, \dots, b-1\}$, so $g(n) = \sum_{i < m} g(\varepsilon_i)$ is a sum of m independent random variables with mean μ_g and range at most R . Hoeffding's inequality [7] gives

$$\#\{n < b^m : |g(n) - \mu_g m| \geq u\} \leq 2b^m \exp\left(-\frac{u^2}{2mR^2}\right),$$

and the lemma follows from $b^m \leq bt$, $m \leq \log_b t + 1$ and $|\mu_g m - \mu_g \log_b t| \leq \mu_g$. $\square$

Proof of Theorem 1.3. The proof is by induction on j . Partial summation converts the sum into an integral over t , and Theorem 2.1 evaluates the integrand in a window about the mean; reversing the order of summation and integration then leaves the reciprocal sum of the targets, one level down in the iteration. Write $c_g = d_g/\varphi(d_g)$; the inductive hypothesis is

$$\sum_{\substack{k \leq Y \\ k \in \mathcal{P}^{(j-1)}}} \frac{1}{k} = c_g^{j-1} \log_{j+1} Y + C_{g,j-1} + O\left(\frac{1}{\log_j Y}\right), \quad (8)$$

which for $j = 1$ is Mertens' theorem, with $C_{g,0}$ the Mertens constant and remainder $O(1/\log Y)$. Put $\mathcal{Q} = \mathcal{P}^{(j-1)}$ and $M(t) = \#\{p \leq t : g(p) \in \mathcal{Q}\}$, so that $\mathcal{P}^{(j)} = \{p : g(p) \in \mathcal{Q}\}$. Partial summation and the bound $M(X) \leq \pi(X) \ll X/\log X$ give

$$\sum_{\substack{p < X \\ p \in \mathcal{P}^{(j)}}} \frac{1}{p} = \frac{M(X)}{X} + \int_2^X \frac{M(t)}{t^2} dt = \int_{t_0}^X \frac{M(t)}{t^2} dt + c_0 + O\left(\frac{1}{\log X}\right),$$

where t_0 is a constant to be chosen and $c_0 = \int_2^{t_0} M(t)t^{-2} dt$ is another. Throughout put $L = \log_b t$, and let

$$H = H(t) = \sigma_g \sqrt{L} \log L, \quad W(t) = [\mu_g L - H, \mu_g L + H].$$

Split $M = M_1 + M_2$ according as $g(p) \in W(t)$ or not.

Outside the window. The count M_2 is small enough that its whole contribution to the integral is bounded. Discarding primality and applying Lemma 4.1 with $u = H - \mu_g$,

$$M_2(t) \leq \#\{n \leq t : |g(n) - \mu_g L| > H\} \ll t \exp(-c(\log L)^2)$$

for some $c = c(b, g) > 0$. Substituting $t = b^L$ and then $L = e^v$,

$$\int_{t_0}^X \frac{M_2(t)}{t^2} dt \ll \int_{t_0}^X e^{-c(\log \log_b t)^2} \frac{dt}{t} \ll \int_1^\infty e^{-c(\log L)^2} dL = \int_0^\infty e^{-cv^2+v} dv = O(1).$$

Inside the window. This is where Theorem 2.1 enters: it evaluates M_1 one target at a time. Enlarging t_0 we may assume $\mu_g L - H > d_g$ for $t \geq t_0$, so every prime $k \in W(t)$ is coprime to d_g . By Brun–Titchmarsh the number of primes in $W(t)$ is $\ll H/\log H \ll \sqrt{L}$. Applying Theorem 2.1 with $\varepsilon = \frac{1}{4}$ to each $k \in \mathcal{Q} \cap W(t)$, and the Siegel–Walfisz theorem to $\pi_k(t)$ for the fixed modulus d_g ,

$$M_1(t) = \frac{c_g \pi(t)}{\sqrt{2\pi\sigma_g^2 L}} \sum_{k \in \mathcal{Q} \cap W(t)} \exp\left(-\frac{(k - \mu_g L)^2}{2\sigma_g^2 L}\right) + O\left(\frac{t}{(\log t)^{5/4}} + \sqrt{L} t e^{-c\sqrt{\log t}}\right).$$

Both error terms are harmless: $\int_{t_0}^X t^{-1}(\log t)^{-5/4} dt = O(1)$, and $\int_{t_0}^X \sqrt{\log t} e^{-c\sqrt{\log t}} t^{-1} dt = O(1)$ after the substitution $\log t = w^2$.

The main term. Each target k will now receive its own integral, which Laplace’s method will show to be close to $1/k$. All terms being nonnegative, we may reverse the order of summation and integration:

$$\int_{t_0}^X \frac{c_g \pi(t)}{t^2 \sqrt{2\pi\sigma_g^2 L}} \sum_{k \in \mathcal{Q} \cap W(t)} e^{-\frac{(k - \mu_g L)^2}{2\sigma_g^2 L}} dt = c_g \sum_{k \in \mathcal{Q}} I_k(X),$$

where $I_k(X)$ is the same integrand integrated over those $t \in [t_0, X]$ with $k \in W(t)$. Substituting $t = b^L$ and $\pi(t) = t(\log t)^{-1}(1 + O((\log t)^{-1}))$,

$$I_k(X) = \int \frac{1}{L \sqrt{2\pi\sigma_g^2 L}} \exp\left(-\frac{(k - \mu_g L)^2}{2\sigma_g^2 L}\right) dL \cdot (1 + O(1/k)),$$

the integral being over an interval of L . Write $L = k/\mu_g + v$. On the range $|v| \ll \sqrt{k} \log k$ allowed by the window one has $L = (k/\mu_g)(1 + O(\mu_g v/k))$, whence

$$\frac{(k - \mu_g L)^2}{2\sigma_g^2 L} = \frac{\mu_g^3 v^2}{2\sigma_g^2 k} \left(1 + O(k^{-1/2} \log k)\right), \quad \frac{1}{L \sqrt{2\pi\sigma_g^2 L}} = \frac{\mu_g^{3/2}}{k^{3/2} \sqrt{2\pi\sigma_g^2}} \left(1 + O(k^{-1/2} \log k)\right).$$

The Gaussian integral $\int_{\mathbb{R}} \exp(-\mu_g^3 v^2 / 2\sigma_g^2 k) dv = (2\pi\sigma_g^2 k / \mu_g^3)^{1/2}$, and the part of it beyond the window is $\ll \sqrt{k} \exp(-\frac{1}{2}(\log L_k)^2)$ with $L_k = k/\mu_g$, which is smaller than any fixed power of $1/k$. Hence, for every k whose peak b^{k/μ_g} lies in $[t_0, X]$ at distance more than $2H$ from either endpoint,

$$I_k(X) = \frac{1}{k} \left(1 + O\left(\frac{(\log k)^3}{\sqrt{k}}\right)\right),$$

while $I_k(X) \ll 1/k$ always. The k excluded by the endpoint condition number $O(\sqrt{\Lambda} \log \Lambda)$ with $\Lambda = \log_b X$, and each contributes $\ll 1/\Lambda$, for a total of $O(\Lambda^{-1/2} \log \Lambda)$. For every k not so excluded, $I_k(X) = I_k(\infty)$, since the constraint $t \leq X$ is then inactive on the range where $k \in W(t)$.

Every error term above was bounded in absolute value by a function integrable on $[t_0, \infty)$, or by the tail of a convergent series; there were four in all. The error term of Theorem 2.1 was controlled by the integral $\int^\infty t^{-1}(\log t)^{-5/4} dt$; the Siegel–Walfisz term by $\int^\infty \sqrt{\log t} e^{-c\sqrt{\log t}} t^{-1} dt$; the contribution from outside the window, through Lemma 4.1, by $\int^\infty e^{-cv^2+v} dv$; and the Laplace approximation by the series $\sum_k k^{-3/2}(\log k)^3$. Each therefore converges as $X \rightarrow \infty$, contributing to a constant $D = D(b, g, j)$, and each remainder is $O((\log X)^{-1/4})$ at worst. Writing $\Lambda = \log_b X$, we obtain

$$\sum_{\substack{p < X \\ p \in \mathcal{P}^{(j)}}} \frac{1}{p} = c_g \sum_{\substack{k \in \mathcal{Q} \\ k \leq \mu_g \Lambda}} \frac{1}{k} + D + O((\log X)^{-1/4}).$$

Applying (8) with $Y = \mu_g \Lambda$ gives the theorem, with $C_{g,j} = c_g C_{g,j-1} + D$: the remainder of (8) becomes $O(1/\log_{j+1} X)$, since $\log \log Y = \log_{j+1} X \cdot (1 + o(1))$ is one logarithm short of $\log \log X$, and $\log_i(\alpha \log X) = \log_{i+1} X + O(1/\log_2 X)$ for fixed $\alpha > 0$ and $i \geq 2$. For $j = 1$ the same computation with Mertens' theorem in the form $\sum_{k \leq Y} 1/k = \log_2 Y + M + O(e^{-c\sqrt{\log Y}})$ gives (2). $\square$

5. HOW OFTEN $g(p)$ IS PRIME: PROOFS OF THEOREMS 1.8, 1.9 AND 1.11

Proof of Theorem 1.8. The upper bound comes from summing Theorem 2.1 over the primes in the window, with Brun–Titchmarsh controlling their number; the second assertion is then extracted from Theorem 1.3 by averaging. Keep the notation $L = \log_b x$, $y = \mu_g L$, $V = \sigma_g^2 L$ and $W = [y - H, y + H]$ with $H = \sigma_g \sqrt{L} \log L$, and note $V/y = \sigma_g^2/\mu_g$, so that $\sqrt{V} \asymp \sqrt{y}$ and $\log y \asymp \log \log x$. By Lemma 4.1 the primes with $g(p) \notin W$ number $\ll x \exp(-c(\log L)^2) = o(\pi(x)/\log y)$, and Theorem 2.1 with $\varepsilon = \frac{1}{4}$ applied to the $\ll \sqrt{L}$ primes $k \in W$ gives

$$M(x) \ll \frac{\pi(x)}{\sqrt{V}} \sum_{\substack{k \text{ prime} \\ k \in W}} e^{-(k-y)^2/(2V)} + O\left(\frac{\pi(x)}{(\log x)^{1/4}}\right).$$

Partition W into blocks of length $\sqrt{V}$. The block at distance between $(i-1)\sqrt{V}$ and $i\sqrt{V}$ from y carries $\ll \sqrt{V}/\log y$ primes, by Brun–Titchmarsh and $\log \sqrt{V} \asymp \log y$, and on it the Gaussian is at most $e^{-(i-1)^2/2}$. Summing over i ,

$$\sum_{\substack{k \text{ prime} \\ k \in W}} e^{-(k-y)^2/(2V)} \ll \frac{\sqrt{V}}{\log y} \sum_{i \geq 1} e^{-(i-1)^2/2} \ll \frac{\sqrt{V}}{\log y},$$

which gives $M(x) \ll \pi(x)/\log y \ll \pi(x)/\log \log x$.

For the second assertion, let ν be the measure $dt/(t \log t \log_2 t)$, so that $\nu([t_0, X]) = \log_3 X + O(1)$, and put $\lambda(t) = M(t) \log_2 t / \pi(t)$, which, by what precedes, is bounded by a constant $B = B(b, g)$. Since $\pi(t)/t^2 = (1 + O(1/\log t))/(t \log t)$, Theorem 1.3 with $j = 1$ reads

$$\int_{t_0}^X \lambda d\nu = \int_{t_0}^X \frac{M(t)}{t^2} dt + O(1) = c_g \log_3 X + O(1),$$

with $c_g = d_g/\varphi(d_g) \geq 1$. If the set where $\lambda \geq c_g/2$ had ν -measure $\eta \log_3 X$, then $\int \lambda d\nu \leq \frac{c_g}{2} \log_3 X + B\eta \log_3 X + O(1)$, forcing $\eta \geq c_g/(2B) + o(1)$. Taking $\delta = \min(c_g/2, c_g/(4B))$ completes the proof. $\square$

For the normal order we need Turán's inequality in intervals of length $\sqrt{y}$, where it is still elementary.

Lemma 5.1. *Let $y \geq 16$ and $\sqrt{y} \leq H \leq y/2$. Then*

$$\sum_{|k-y| \leq H} (\omega(k) - \log_2 y)^2 \ll H \log_2 y.$$

Proof. We truncate ω and compute two moments, as in Turán's original argument. Put $z = y^{1/4}$ and $\omega_z(k) = \#\{\ell \leq z : \ell \text{ prime, } \ell \mid k\}$, so that $0 \leq \omega(k) - \omega_z(k) \leq 4$ for $k \leq 2y$. Counting multiples in the interval $I = [y - H, y + H]$,

$$\sum_{k \in I} \omega_z(k) = \sum_{\ell \leq z} \left(\frac{2H}{\ell} + O(1) \right) = 2H \log_2 z + O(H + \pi(z)),$$

and $\pi(z) \ll y^{1/4} \ll H$. Similarly, separating $\ell_1 = \ell_2$ from $\ell_1 \neq \ell_2$,

$$\sum_{k \in I} \omega_z(k)^2 = 2H(\log_2 z)^2 + O(H \log_2 z + \pi(z)^2),$$

and $\pi(z)^2 \ll \sqrt{y}/(\log y)^2 \ll H$. Expanding the square and using $\log_2 z = \log_2 y + O(1)$ gives the lemma. $\square$

Proof of Theorem 1.9. The plan is to confine $g(p)$ to a short window about y and to apply Lemma 5.1 there. Let $\epsilon > 0$ and $A \geq A_0 := \max(1, \sqrt{\mu_g}/\sigma_g)$; the lower bound ensures $A\sqrt{V} \geq \sqrt{y}$, as Lemma 5.1 requires. Keep the notation above, and write $W_A = [y - A\sqrt{V}, y + A\sqrt{V}]$. Summing Theorem 2.1 over the integers $k \notin W_A$ inside W , and using Lemma 4.1 outside W , the primes $p \leq x$ with $g(p) \notin W_A$ number

$$\ll \frac{\pi(x)}{\sqrt{V}} \sum_{|k-y| > A\sqrt{V}} e^{-(k-y)^2/(2V)} + O\left(\frac{\pi(x) \log \log x}{(\log x)^{1/4}}\right) + o(\pi(x)) \ll e^{-A^2/2} \pi(x) + o(\pi(x)),$$

the second error term arising because $W \setminus W_A$ contains $\asymp \sqrt{L} \log L$ integers, of which only $\ll \sqrt{L}$ are prime. By Lemma 5.1 applied with $H = A\sqrt{V}$, the integers $k \in W_A$ with $|\omega(k) - \log_2 y| > \epsilon \log_2 y$ number $\ll_\epsilon A\sqrt{V}/\log_2 y$, and by Theorem 2.1 again the primes p with $g(p)$ among them number

$$\ll_\epsilon \frac{\pi(x)}{\sqrt{V}} \cdot \frac{A\sqrt{V}}{\log_2 y} + A\sqrt{V} \cdot \frac{\pi(x)}{(\log x)^{3/4}} = o(\pi(x))$$

for fixed A and ϵ , since $\log_2 y \rightarrow \infty$ and $\sqrt{V} \ll (\log x)^{1/2}$. Hence all but $O(e^{-A^2/2} \pi(x)) + o(\pi(x))$ primes $p \leq x$ satisfy $|\omega(g(p)) - \log_2 y| \leq \epsilon \log_2 y$. Letting $A \rightarrow \infty$ and then $\epsilon \rightarrow 0$, and noting that $\log_2 y = \log_3 x + O(1/\log_2 x)$ while $\log_3 p = \log_3 x + O(1/\log_2 x)$ for all but $o(\pi(x))$ of the primes $p \leq x$, the assertion for ω follows. For Ω it suffices to add that $\sum_{k \in I} (\Omega(k) - \omega(k)) \ll H$ for I as in Lemma 5.1, since the prime powers $\ell^a \leq 2y$ with $a \geq 2$ number $\ll \sqrt{y} \ll H$ and contribute $\ll H \sum_{a \geq 2} \sum_{\ell} \ell^{-a} + \sqrt{y}$. $\square$

Proof of Theorem 1.11. We rerun the proof of the upper bound in Theorem 1.8; where Brun–Titchmarsh could only bound the number of primes in a block, Hypothesis 1.10 now evaluates it. Put $L = \log_b x$, $y = \mu_g L$, $V = \sigma_g^2 L$, $H = \sigma_g \sqrt{L} \log L$ and $W = [y - H, y + H]$. By Lemma 4.1 the primes $p \leq x$ with $g(p) \notin W$ number $\ll x \exp(-c(\log L)^2) = o(\pi(x)/\log y)$. Exactly as in the proof of Theorem 1.3, Theorem 2.1 with $\varepsilon = \frac{1}{4}$ and the Siegel–Walfisz theorem give

$$\#\{p \leq x : g(p) \text{ prime}\} = \frac{c_g \pi(x)}{\sqrt{2\pi V}} \sum_{\substack{k \text{ prime} \\ k \in W}} e^{-(k-y)^2/(2V)} + o\left(\frac{\pi(x)}{\log y}\right),$$

the error term absorbing $O(\sqrt{L} \pi(x)(\log x)^{-3/4})$. Fix $A > 0$ and $c > 0$ and partition $[y - A\sqrt{V}, y + A\sqrt{V}]$ into blocks of length $c\sqrt{V}$. Since $V/y = \sigma_g^2/\mu_g$ exactly, a block of length $c\sqrt{V}$ has length $c'\sqrt{y'}(1 + o(1))$ with $c' = c\sigma_g/\sqrt{\mu_g}$ for each of its endpoints y' , so Hypothesis 1.10 applied at $c'(1 \pm \delta)$ and the monotonicity of π sandwich the count of primes in the block between $(1 \pm 2\eta)(1 + o(1))c\sqrt{V}/\log y$ for any fixed $\eta > 0$. Letting $\eta \rightarrow 0$, each block supplies $(1 + o(1))c\sqrt{V}/\log y$ primes, while the Gaussian varies across a block by a factor $1 + O(Ac)$. Summing over blocks, letting $c \rightarrow 0$ and then $A \rightarrow \infty$, and bounding the range $A\sqrt{V} < |k - y| \leq H$ by Brun–Titchmarsh at $\ll \sqrt{V}e^{-A^2/2}/\log y$,

$$\sum_{\substack{k \text{ prime} \\ k \in W}} e^{-(k-y)^2/(2V)} = (1 + o(1)) \frac{1}{\log y} \int_{\mathbb{R}} e^{-u^2/(2V)} du = (1 + o(1)) \frac{\sqrt{2\pi V}}{\log y}.$$

Since $\log y = \log(\mu_g \log_b x)$, the theorem follows. $\square$

6. COMPLEMENTS

6.1. The modulus d_g . Let ℓ be a prime and $e \geq 1$. From (4), $\ell^e \mid d_g$ if and only if $\ell^e \mid b - 1$ and $g(a) \equiv ag(1) \pmod{\ell^e}$ for $0 \leq a < b$. When g is given by a polynomial $P \in \mathbb{Z}[X]$ and $e = 1$, the second condition says that the function $a \mapsto P(a) - aP(1)$ vanishes on $\mathbb{F}_\ell$, that is, that $X^\ell - X$ divides $P(X) - P(1)X$ in $\mathbb{F}_\ell[X]$; here $\ell \leq b - 1$, so the digits do cover $\mathbb{F}_\ell$.

For power sums this yields a formula in every base. For $g(a) = a^r$ one has $g(1) = 1$, and the condition at ℓ becomes $a^r \equiv a \pmod{\ell}$ for all a , which holds precisely when $\ell - 1 \mid r - 1$. No square of a prime divides d_g once $r \geq 2$, since $\ell^2 \nmid \ell^r - \ell$. Hence

$$d_g = \prod_{\substack{\ell \mid b-1 \\ \ell-1 \mid r-1}} \ell \quad (r \geq 2), \quad d_g = b - 1 \quad (r = 1).$$

In base ten this is 9 for $r = 1$, 3 for odd $r \geq 3$ and 1 for even r , corresponding to the congruences $s(n) \equiv n \pmod{9}$ and $\sum_j \varepsilon_j(n)^r \equiv n \pmod{3}$ for odd r , and to the absence of any congruence in the even case. In base three the formula gives 2 for every r , reflecting $a^r \equiv a \pmod{2}$. Thus S is the most accommodating of these functions: $d_S = \gcd(2, 6, 12, 20, 30, 42, 56, 72, 9) = 1$, the congruence in Theorem 2.1 is vacuous, and every sufficiently large integer occurs as $S(p)$ for some prime p . The restriction $r \geq 2$ is needed for the squarefreeness: in general d_g can be divisible by a square, as $P(X) = X$ in base ten shows.

6.2. Sharpness of the hypotheses. Theorem 2.1 is useful exactly for targets in the window

$$|k - \mu_g \log_b x| \leq \sigma_g \sqrt{(1 - 2\varepsilon) \log_b x \log \log x},$$

which is where the Gaussian in (5) exceeds the error term. This is narrower than the window used in the proof of Theorem 1.3, where only an aggregate bound is wanted. Outside the window the theorem asserts nothing about a single k , and each of the ways in which our hypotheses can fail places the available targets there. Suppose first that $\delta = \gcd(g(1), \dots, g(b-1)) > 1$. Then $\delta \mid g(n)$ for every n , so a prime value forces $g(p) = \delta$; when $\mu_g > 0$ this is a fixed target at distance $\asymp \log x$ from the centre. Suppose next that $\mu_g < 0$. Then every positive target lies at that distance. Finally, with the hypotheses in force but $\mu_g > 0$, a fixed k is again out of reach. In each case the Gaussian contributes x^{-c} against an error term of size $\pi(x)/\log x$, and nothing survives.

Clause (ii) of Theorem 1.1 is the one case where fixed targets sit at the centre of the window rather than in its tail. There the coprimality hypothesis (3) can be dispensed with when δ is prime: the assertion that $g(p) = \delta$ becomes $h(p) = 1$ for $h = g/\delta$, and h satisfies (3) with

$\mu_h = 0$. For an instance of the clause, take $b = 10$ with $g(0) = 0$ and $g(a) = a - 5$ for $1 \leq a \leq 9$; then $\mu_g = 0$, $\sigma_g^2 = 6$, (3) holds and $d_g = 1$. Since the proof of clause (ii) uses only $\gcd(k, d_g) = 1$, every integer k , in particular every prime, is the value $g(p)$ for $\gg \pi(x)(\log x)^{-1/2}$ primes $p \leq x$.

The words “sufficiently large” in clause (i) of Theorem 1.1 cannot be dropped. The value $k = 3$ is prime and is never attained: $S(n) = 3$ forces n to have exactly three digits equal to 1 and the rest equal to 0, so that $s(n) = 3$ and $3 \mid n$. Likewise $S(n) = 1$ forces $n = 10^j$, so that 1 and 3 are omitted from the set of values $S(p)$. Congruence obstructions can therefore survive at small values even when $d_g = 1$. There is no contradiction: d_g governs g on all of $\mathbb{N}$, whereas $S(n) = k$ with k small confines n to the sparse set of integers with at most k nonzero digits, where s and S are tightly linked.

6.3. The function S . On $\{0, 1\}$ one has $a^2 = a$, so in base 2 the function S coincides with the digit sum and Corollary 1.4 reduces to the theorem of Drmota, Mauduit and Rivat. In base 10 the general theory is needed, S being no function of s : $s(19) = s(28)$ while $S(19) = 82$ and $S(28) = 68$.

For $b = 10$ and $g = S$, the iteration in Corollary 1.6 cannot be continued indefinitely. The map S is the one that defines the happy numbers: every orbit reaches 1 or enters the cycle

$$4, 16, 37, 58, 89, 145, 42, 20.$$

The orbit of a fixed n is eventually periodic, and neither $\{1\}$ nor the eight-term cycle consists of primes, so no prime has all of its S -iterates prime: the restriction to finitely many steps in Corollary 1.6 is not an artifact of the method.

6.4. What sieve methods give. One may try to detect prime values of $S(p)$ by a sieve, the values having size of order $\log x$. The equidistribution of $S(p)$ modulo m furnished by [10] is uniform for m up to roughly $\sqrt{\log x / \log \log x}$, which permits inclusion and exclusion over the squarefree moduli composed of primes below $c \log \log x$ for small enough c . A routine argument then gives infinitely many primes p with $S(p)$ free of prime factors below $c \log \log p$. This does not improve on Theorem 1.9; what it produces is a thin set of p on which the *smallest* prime factor of $S(p)$ is atypically large. Detecting primality would require sifting up to $S^{1/2}$, hence moduli of size $\exp((\log x)^{1/2})$, beyond the available uniformity by an exponential; and the parity obstruction would remain in any case.

6.5. Two numerical remarks. Neither Theorem 1.3 nor Theorem 1.9 is visible in its leading term at an accessible height. Indeed $\log_3 X$ is only 1.07 at $X = 10^8$, while the secondary term $\log(1 + \kappa_g / \log_2 X)$ in (2), taking $g = S$ and $\kappa_S = \log \frac{57}{2 \log 10} = 2.516$, is worth 0.62 there, more than half the size of the leading term. The constant, on the other hand, is visible. For $g = S$ the difference between $\sum_{p < X, S(p) \text{ prime}} 1/p$ and $\log(\log_2 X + \kappa_S)$ moves by only 0.004 between $X = 10^4$ and $X = 10^8$, so the limit whose existence (2) asserts is already in evidence; extrapolation gives $C_{S,1} = -1.19$.

So is the mechanism behind Theorem 1.9. Over the primes of eight decimal digits the mean of $\omega(S(p))$ is 2.055, nearly twice $\log_3(10^8)$, so the normal order itself is nowhere in sight. But the proof ran through the window, and the window makes a prediction of its own: weighting the integers k by the Gaussian of (5) gives 2.051 for the mean of $\omega(k)$, and the two variances are 0.480. It is the window, not the iterated logarithm, that the computation sees at this height.

6.6. Further questions. A weight may be allowed to depend on the position of the digit, or on several consecutive digits at once, as in $\sum_j F(\varepsilon_{j+1}(n), \varepsilon_j(n))$. The second of these is the setting of automatic sequences, where the equidistribution of the values along the primes is known but a local limit theorem, as far as we are aware, is not. One may also ask for two

digital conditions at once, say that $s(p)$ and $S(p)$ be simultaneously prime; the nearest available model is the work of Drmota, Mauduit and Rivat on primes in two bases [4].

Finally, the results above are ineffective, the constant implied in Theorem 2.1 not being explicit; Lehmann [8] has done the corresponding work for the digit sum, and the general case would require the constants of [11] to be tracked.

USE OF AUTOMATED TOOLS

Most of this manuscript was written by large language models, principally Opus 5 and Fable 5, working under the author's direction; the exposition was afterwards revised by the author and the same models. The numerical values quoted in §6.5 were computed by machine, and the scripts that produce them are distributed with the paper in the companion repository

<https://github.com/vibefrtz/vibemath>

(folder `prime-values-digital-functions`, archived at [doi:10.5281/zenodo.22093372](https://doi.org/10.5281/zenodo.22093372)).

A Lean 4 formalisation accompanies the paper and was written the same way. It derives Theorem 1.1 in the case $d_g = 1$, and with it Corollaries 1.4 and 1.5 in that case, from a single axiom transcribing Theorem 2.1; Lemma 2.2 and the assertions of §6.1, §6.2 and §6.3 are proved outright, on no hypothesis at all. Theorems 1.2, 1.3, 1.8, 1.9 and 1.11 are not formalised, nor is clause (ii) of Theorem 1.1. The formalisation compiles against Mathlib with no incomplete proofs, and the axiom it assumes has been compared clause by clause with the statement of Theorem 2.1 in its source.

Responsibility for the mathematics rests with the author.

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